

Fulton's *Algebraic Curves*

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§ 1 Affine Algebraic Sets

1.1 Algebraic Preliminaries

1.4: For k infinite, the only polynomial that is zero everywhere is the zero polynomial.

Proof. Let $F \in k[X_1, \dots, X_n]$ be a polynomial such that $F(a_1, \dots, a_n) = 0$ for all $a_1, \dots, a_n \in k$. We proceed by induction on the number of variables n . For the case $n = 0$ the statement holds trivially. Now suppose the statement holds for $n - 1$. Then we may write $F = \sum F_i X_n^i$ for $F_i \in k[X_1, \dots, X_{n-1}]$. Now suppose that one of the F_i is not equal to zero. Then there are only finitely many roots for the polynomial $F(a_1, \dots, a_{n-1}, X_n)$, as it is univariate. However the field k is infinite so this polynomial cannot possibly vanish everywhere on k . Therefore $F = 0$. $\triangle$

1.2 Affine Space and Algebraic Sets

1.3 The Ideal of a Set of Points

1.21: An ideal generated by linear polynomials is maximal and the quotient is isomorphic to the field of coefficients

Proof. Suppose $I = (X_1 - a_1, \dots, X_n - a_n)$, and let $J \supseteq I$. Then I is the ideal of the point $\mathbf{a} = (a_1, \dots, a_n)$ and $V(J) \subseteq V(I)$. However $V(I) = \{\mathbf{a}\}$, so $J = I$. That is, I is maximal. Now take the quotient $k[X_1, \dots, X_n]/I$. Now the only ideals in the quotient are the whole ring and the zero ideal, which implies that the ring is a field. The only field contained in $k[X_1, \dots, X_n]$ is k , so the quotient must be isomorphic to this field. $\triangle$

Proposition 1: An algebraic set is irreducible if and only if its ideal is prime.

Proof. First the backward direction. Suppose $I(V)$ not prime. Then there is $fg \in I$ such that $f \notin I$ and $g \notin I$. Then $V(f) \cap V \cup V(g) \cap V = V$, and so V is reducible.

Now the forward direction. Suppose $V = U \cup W$ is reducible. Then for $f \in I(U) \setminus I$ and $g \in I(W) \setminus I$ we have that $fg \in I$, so I is not prime. $\triangle$

1.4 The Hilbert Basis Theorem

1.22: Bijections between ideals of R/I and ideals containing I .

Proof. Part (a) asks us to prove the above statement for any ideals. First the forward direction. If J' is an ideal of R/I then $\pi^{-1}(J')$ is in particular closed under multiplication and addition of generators by the fact that π is a homomorphism. That is, it is an ideal. The fact that it contains I follows by the fact that $0 \in J'$ and $\pi^{-1}(0) = I$.

Part (b) asks us to prove that radicality, primality, and maximality are preserved. First radicality. Supposing J' is radical, we have $f^m \in J'$ implies that $f \in J'$. Then taking preimages, $\pi^{-1}(f^m) = f^m + I = (f + I)^m = \pi^{-1}(f)$, so $\pi^{-1}(J')$ is radical. Likewise, if J is radical then $\pi(J)$ is obviously radical because $\pi(f^m) = \pi(f)^m$ and $\pi(f) \in \pi(J)$. Now primality. If J' is prime then $\pi^{-1}(J') = J$ is because $\pi^{-1}(fg) = fg + I = (f + I)(g + I)$ and one of $f + I$ or $g + I$ is in the preimage. Conversely, if J is prime then multiplicativity of the homomorphism shows that J' is as well. For maximality, note that projection preserves containment.

For part (c), note that if J is finitely generated then we may write any element of J' as

$$\pi(x) = \pi\left(\sum c_\alpha y_i\right) = \sum \pi(c_\alpha) \pi(y_i)^\alpha$$

for y_i generators of J . $\triangle$

1.5 Irreducible Components of an Algebraic Set

1.6 Algebraic Subsets of the Plane

1.7 Hilbert's Nullstellensatz

1.38: Bijections between algebraic subsets of an algebraic set $V = V(I)$ and radical ideals of $k[X_1, \dots, X_n]/I$.

Proof. The Nullstellensatz demonstrates bijections between algebraic sets and radical ideals, and 1.22(a-b) (above) demonstrates the necessary bijections once one notes the inclusion-reversing properties of $I(\cdot)$ and $V(\cdot)$ as functors (J contains I exactly when $V(J)$ is a subset of $V(I)$) and recalls that irreducible varieties have prime ideals, while points have maximal ideals. $\triangle$

1.8 Modules: Finiteness Conditions

1.9 Integral Elements

1.10 Field Extensions

§ 2 Affine Varieties

2.1 Coordinate Rings

2.1: Polynomials mapped to polynomial functions give a ring hom with kernel $I(V)$.

Proof. Let $\varphi : k[X_1, \dots, X_n] \rightarrow \mathcal{F}(V, k)$ assign each polynomial a polynomial function. This is obviously a ring homomorphism, and the kernel is shown to be $I(V)$ by noting that $f \rightarrow 0 \iff f(v) = 0$ for all $v \in V \iff f \in I(V)$. $\triangle$

2.2: Bijections between algebraic subsets and radical ideals of $\Gamma(V)$.

Proof. This follows by recalling that $\Gamma(V) = k[X_1, \dots, X_n]/I(V)$ and applying 1.22 and 1.38 (above). $\triangle$

2.3: Subvarieties give maps on $\Gamma(V)$

Proof. For part (a), note that $I_V(W)$ is the image of $I(W)$ under the quotient $k[X_1, \dots, X_n]/I(V)$. Then obviously any polynomial on V restricts to one on W by the fact that polynomials are still polynomials under the quotient map. This map is surjective precisely because it is a quotient map $k[X_1, \dots, X_n]/I(V) \rightarrow k[X_1, \dots, X_n]/I(W)$ and $I(W) \supseteq I(V)$. The isomorphism follows because $I_V(W)$ is approximately $I(W)/I(V)$. $\triangle$

2.4: Points have coordinate ring k and finite dimensional coordinate rings.

Proof. The equivalence of (ii) and (iii) are the content of 1.21 (above). Corollary 4 of the Nullstellensatz tells us that V is finite, and because it is a variety it is irreducible and hence a point. (ii) implies (iii) straightforwardly as k is a k -vector space of dimension one. $\triangle$

2.2 Polynomial Maps

2.6: Composition of polynomial maps is a polynomial map.

Proof. Let $\varphi : V \rightarrow W$ and $\psi : W \rightarrow Z$. Now $\psi \circ \varphi(F) = F \circ \psi \circ \varphi = \tilde{\varphi}(F \circ \psi) = (\tilde{\varphi} \circ \tilde{\psi})(F)$. If φ and ψ are polynomial maps, then $\varphi = (T_1, \dots, T_n)$ and $\psi = (S_1, \dots, S_m)$, and from this it follows that $\psi \circ \varphi = (S_1(T_1, \dots, T_n), \dots, S_m(T_1, \dots, T_n))$. Now one can see that $\psi \circ \varphi$ is obviously polynomial by expanding. $\triangle$

2.7: Preimages under polynomial maps preserve algebraicity and irreducibility.

Proof. Let $\varphi : V \rightarrow W$ be a polynomial map and X an algebraic subset of W . Let $I_W(X)$ be the ideal of X in $\Gamma(W)$ and take $Y = V(F \in \Gamma(V) \mid F \circ \varphi \in I_W(X))$. Then $\varphi(Y) = X$. The claim on irreducibility follows by the fact that if Y is reducible then there will be some $F, G \in \Gamma(V)$ such that $FG \circ \varphi \in I_W(X)$ but neither will have pullback lying in the ideal, implying that it is not prime and so X is reducible. $\triangle$

2.9: Polynomial maps and subvarieties.

Proof. Let $\varphi : V \rightarrow W$ be a polynomial map and $V' \subseteq V$, $W' \subseteq W$ subvarieties and $\varphi(V') \subseteq W'$. For part (a), we have that because $\varphi(V') \subseteq W'$, $V' \subseteq \varphi^{-1}(W')$. Then $\varphi^{-1}(W')$ is a variety by 2.7, and so we conclude that $I_V(V') \supseteq I_V(\varphi^{-1}(W'))$. Now note that any polynomial F vanishing on $\varphi^{-1}(W')$ satisfies $\tilde{\varphi}(F) \in I_W(W')$. Therefore $I_W(W') \subseteq I_V(\varphi^{-1}(W'))$ and the result holds.

Part (b) seems more or less obvious: $\varphi(V') \subseteq W'$ means that the map is defined on all of V' , and it being polynomial at every point in V means that it is, in particular, polynomial at $V' \subseteq V$. $\triangle$

2.10: Projection map is polynomial

Proof. This is the map $\varphi = (x_1, \dots, x_r)$. $\triangle$

2.3 Coordinate Changes

2.14: Linear subvarieties

Proof. Let $V \subseteq \mathbb{A}^n(k)$ be a linear subvariety (that is, $V = V(F_1, \dots, F_r)$ for F_i linear) and let $I = I(V)$. First we show part (a). Let T be an affine change of coordinates on $\mathbb{A}^n$. Now $V^T = V(F_1^T, \dots, F_r^T) = V(F_i(T_1, \dots, T_n) \mid 1 \leq i \leq r)$. Writing $F = \sum a_i X_i$ by linearity, we now have $F^T = \sum a_i T_i = \sum a_i \sum b_j X_j$ which is obviously a linear polynomial. Implicitly we are using the fact that I^T is generated by F_i^T , which is clear. Therefore V^T is linear.

For part (b) we proceed by induction on r . For $r = 1$ and $V = V(F)$, write $F = \sum a_i X_i$. Let j be the lowest index for which $a_j \neq 0$ so we may write $T_j = (X_j - \sum_{i \neq j} a_i X_i)/a_j$ and $T_i = X_i$ for $i \neq j$. Then $F^T = F(T_1, \dots, T_n) = \sum_{i \neq j} a_i X_i + a_j(X_j - \sum_{i \neq j} a_i X_i)/a_j = X_j$. Now consider S for which $S_j = X_n$ and $S_i = X_i$ for all other $i \neq j$. Now observe that $F^{T \circ S} = F(T_1(S_1, \dots, S_n), \dots, T_n(S_1, \dots, S_n)) = (F^T)(S_1, \dots, S_n) = (F^T)^S = X_j^S = X_n$. Then the base case of induction holds. Now suppose that for $r - 1$ the statement holds and consider $V = V(F_1, \dots, F_r)$. Now $V(F_1, \dots, F_{r-1})^T = V(X_{m+1}, \dots, X_n)$ for some m and some affine coordinate change T . Then F_r^T is a polynomial in $X_{m+1}, \dots, X_n$ and we may apply the above construction to produce a map S that sends F^T to X_m and fixes all other variables. Postcomposition of T with this map then yields the desired map.

For part (c), we suppose for the sake of contradiction that $T : X = V(X_{m+1}, \dots, X_n) \rightarrow Y = V(X_{s+1}, \dots, X_n)$ is an isomorphism for $m < s$. Let $I = (X_{m+1}, \dots, X_n)$ so that $I^T = (X_{s+1}, \dots, X_n)$. Now the induced map on coordinate rings $\tilde{T} : \Gamma(Y) \rightarrow \Gamma(X)$ is an isomorphism. However note that $I(X) = (X_{m+1}, \dots, X_n)$ and $I(Y) = (X_{s+1}, \dots, X_n)$, so the dimensions of the two coordinate rings differ as k vector spaces. This implies that there cannot be a bijection between the two, so in particular they are not isomorphic. $\triangle$

2.4 Rational Functions and Local Rings

2.18: Bijection between subvarieties containing P and prime ideals in $\mathcal{O}_P(V)$.

Proof. Suppose I is prime in $\mathcal{O}_P(V)$ so that $I \cap \Gamma(V)$ is also prime ($fg \in I \cap \Gamma(V)$ implies that wlog $f \in I$ but $fg \in I \cap \Gamma(V)$ implies wlog f is polynomial and clears g 's denominator). By the proof of Proposition 3 we have that $I \cap \Gamma(V)$ generates I . Now apply the bijection between prime ideals in $\Gamma(V)$ and subvarieties so that $I \cap \Gamma(V)$ gives a subvariety, and note that $P \in V(I \cap \Gamma(V))$ by the fact that every $h \in I$ is defined at P . $\triangle$

2.21: Polynomial maps extend to homomorphisms on local rings.

Proof. Suppose $\varphi : V \rightarrow W$ is a polynomial map such that $P \in V$ has $\varphi(P) = Q$. Now consider the map on coordinate rings $\tilde{\varphi} : \Gamma(W) \rightarrow \Gamma(V)$. We wish to show that $\tilde{\varphi}$ extends uniquely to a homomorphism $\mathcal{O}_Q(W) \rightarrow \mathcal{O}_P(V)$. We define such a homomorphism by $\tilde{\varphi}(f/g) = \tilde{\varphi}(f)/\tilde{\varphi}(g)$. This map respects the equivalence relations as if $f/g \sim p/q$ then $fq - pg = 0$ and so $\tilde{\varphi}(fq - pg) = \tilde{\varphi}(f)\tilde{\varphi}(q) - \tilde{\varphi}(g)\tilde{\varphi}(p) = 0$ and obviously restricts to the map on coordinate rings. The fact that it is a homomorphism follows by the fact that it is a homomorphism on coordinate rings and the definition of the field of fractions. Uniqueness is clear by construction. In particular, $\tilde{\varphi}(\mathfrak{m}_Q(W)) \subseteq \mathfrak{m}_P(V)$ as if $f/g \in \mathfrak{m}_Q(W)$ vanishes at Q then in particular $f(Q) = 0$ implying that $f(\varphi(P)) = 0$ and so $\tilde{\varphi}(f)(P) = 0$. $\triangle$

2.22: Isomorphisms on local rings.

Proof. Suppose $T : \mathbb{A}^n \rightarrow \mathbb{A}^n$ is an affine change of coordinate with $T(P) = Q$. Notably the coordinate rings are isomorphic and so by construction of $\tilde{T}$ the local rings are isomorphic as well. If V is a subvariety of $\mathbb{A}^n$ such that $P \in V^T$, then the restriction to $\mathcal{O}_Q(V)$ has image given by F^T/G^T for all $F/G \in \mathcal{O}_Q(V)$. However this is precisely the field of fractions of V^T , so long as $P \in V^T$ to ensure that we restrict to where F^T is defined. $\triangle$

2.5 Discrete Valuation Rings

2.23: Order function is independent of uniformizing parameter

Proof. If t is a uniformizing parameter for R then any other uniformizing parameter is of the form $s = ut$ for $u \in R^\times$ a unit. Then if the order of $z \in K = \text{Frac}(R)$ is n we have $z = vt^n$ for some $v \in R^\times$. We may then write $z = vt^n = v(u^{-1}s)^n = ws^n$ for $w = vu^{-n}$ a unit. This expression in terms of the uniformizing parameter is unique, and so the order is fixed. $\triangle$

2.24: The affine line has local ring a DVR.

Proof. First we show part (a). Let $f/g \in \mathfrak{m}_a(V) \subset \mathcal{O}_a(\mathbb{A}_k^1)$. Then f/g vanishes at a , implying that $f = h \cdot (X - a)$ for some $h \in \Gamma(V)$, and we may write $f = h \cdot (X - a)^n$ with h not divisible by $(X - a)$. In this case f/g permits an expression of the form $f/g = h/g \cdot (X - a)^n$, and h/g is a unit because neither is divisible by $(X - a)$. Importantly this unit is unique by construction, and so $\mathcal{O}_a(V)$ is a DVR.

Now part (b). We let $\mathcal{O}_\infty(V) = \{f/g \mid \deg(g) \geq \deg(f)\}$. Let $f/g \in \mathcal{O}_\infty(V)$ and suppose that we take a representative with such that X is not a factor of both f and g . Note that f/g is a unit if and only if $\deg f = \deg g$, as then $g/f \in \mathcal{O}_\infty(V)$. Now for any f/g we may take $X^n f/g$ to be a unit for n sufficiently large, implying that $f/g = X^n f/g \cdot (1/X)^n$. This expression is unique by construction, and so $\mathcal{O}_\infty(V)$ is a DVR. $\triangle$

2.26: DVRs and quotient fields

Proof. Part (a): Let $z \in K$ such that $z \notin R$. Then the order of z is negative, implying that $z = ut^n$ for $n < 0$. Now take an inverse to see that $z^{-1} = u^{-1}t^{-n}$, so in particular z has positive order implying that $z \in \mathfrak{m}$.

Part (b): Suppose $\mathfrak{m}_R \subseteq \mathfrak{m}_S$. Suppose $z \in S - R$. Then $z^{-1} \in \mathfrak{m}_R$ by (a), so $z^{-1} \in \mathfrak{m}_S$. However $z \in S$ so $zz^{-1} \in \mathfrak{m}_S$ which is a contradiction. Therefore $S = R$. $\triangle$

2.28: Valuations give DVRs.

Proof. Let φ be a valuation on K . It is clear that R is a ring and $\mathfrak{m}$ is an ideal. Observe that $\mathfrak{m}$ is the set of non-units in R , and so R has a unique maximal ideal containing every proper ideal of R by the Lemma in 2.4, which is equal to $\mathfrak{m}$. That is, R is local. Suppose $x \in R$. We wish to show that $x = ut^n$ for some irreducible $t \in R$, some unit $u \in R^\times$, and some $n \geq 0$. Fix some irreducible $t \in \mathfrak{m}$ of minimal degree. Then any nonzero $y \in R$ has $0 \leq \varphi(yt^{-m}) < \varphi(t)$ for some m . This implies that yt^{-m} is a unit, as $\varphi(t)$ is minimal. Equivalently, $y = ut^m$ and this is a unique expression by construction.

The converse is obvious. $\triangle$

2.29: Properties of valuations

Proof. Part (a): If $\varphi(a) < \varphi(b)$ then $\varphi(a+b) \geq \min(\varphi(a), \varphi(b)) = \varphi(a)$.

Part (b): If there exists i such that $\varphi(a_i) < \varphi(a_j)$ for all $j \neq i$ then $\varphi(a_1 + \dots + a_n) = \varphi(a_i) \neq \infty$, so $a_1 + \dots + a_n \neq 0$. $\triangle$

2.30: Properties of the quotient $R/\mathfrak{m}$

Proof. Part (a): Let $z \in R$. If $z \in \mathfrak{m}$ then $\lambda = 0$. Suppose $z \notin R$ and note that the coset $z + \mathfrak{m} \in k$ has additive inverse $-\lambda$ such that $z - \lambda = 0 \in k$. However $z - \lambda = 0 \in k$ implies that the preimage lives in $\mathfrak{m}$.

Part (b): **Incomplete** $\triangle$

2.6 Forms

2.35: Forms as vector spaces over k .

Proof. Part (a): A monomial of degree d in $R[X, Y]$ is of the form $X^i Y^j$ such that $i + j = d$. There are $d + 1$ pairs satisfying this, as can be seen by noting that $i = d - j$ is a solution for all $0 \leq j \leq d$. Similarly, in $R[X, Y, Z]$ we have as many monomials as triples $i + j + k = d$ and writing $i = d - j - k$ we see that the number of such triples is given by $\sum_{j=0}^d \sum_{k=0}^i 1 = \sum_{j=0}^d j = (d + 1)(d + 2)/2$.

Part (b) is a consequence of the classical fact that polynomials over a field of finite degree form a vector space by homogenizing.

Part (c): **Incomplete** $\triangle$

2.7 Direct Products of Rings

2.38: Dimensions of ring direct products.

Proof. The direct product is obviously generated as a ring by the generators of each R_i , and so there are $\dim R_i$ generators for each component. Components are independent, so there are $\sum \dim R_i$ generators. Equivalently, the dimension of the product is the sum of dimensions. $\triangle$

2.8 Operations with Ideals

2.39: Ideal sum and product.

Proof. Part (a): $(I_1 + I_2)J = \{a + b \mid a \in I_1, b \in I_2\} \cdot J = \{(a + b)c \mid a \in I_1, b \in I_2, c \in J\} = \{ac + bc \mid \dots\} = I_1J + I_2J$.

Part (b): $(I_1 \dots I_m)^n = (\{a_1 \dots a_m \mid a_i \in I_i\})^n = \{a_1^n \dots a_m^n \mid a_i \in I_i\} = I_1^n \dots I_m^n$. $\triangle$

2.40: Sums and products of comaximal ideals are comaximal.

Proof. Part (a): If I, J are comaximal then $I + J = R$. Then $1 = a + b$ for $a \in I, b \in J$. Now $1 = 1^2 = a^2 + 2ab + b^2 = a(a + 2b) + b^2$. Now $a(a + 2b) \in I$ and $b^2 \in J^2$, so in particular $I + J^2 = R$. This naturally generalizes: $1 = 1^m = af + b^m$ for $f = \sum_{i+j=m} a^i b^j$ where the sum ranges over $0 \leq i < j \leq m - 1$. This shows that I and J^m are radical, and because our rings are commutative $I + J = J + I$ so we may perform the same process to obtain that $J^m + I^n = R$.

Part (b): The Lemma states that I, J comaximal implies that $IJ = I \cap J$, and part (a) shows that any powers of I and J are comaximal. Applying this to $J_i = \bigcap_{j \neq i} I_j$ we note that $I_1^n \cap \bigcap_{j=2}^m I_j^n = I_1^n \cdot \bigcap_{j=2}^m I_j^n = \dots = I_1^n \dots I_m^n = (I_1 \dots I_m)^n = (I_1 \cap \dots \cap I_m)^n$. $\triangle$

2.41: Radical ideals contain powers of ideals (ChatGPT solved this one).

Proof. Suppose $I \subseteq \sqrt{J}$. Then for any $a \in I$ there is some minimal $n \geq 0$ such that $a^n \in J$. For $g_1, \dots, g_m$ generators of I we write $a = \sum b_i g_i$. Then $a^n = \sum b_1^{i_1} \dots b_m^{i_m} g_1^{i_1} \dots g_m^{i_m}$ where the sum is over all $i_1, \dots, i_m$ such that $\sum i_j = n$. Let m_i be the minimal power of g_i such that $g_i^{m_i} \in J$. Taking $n = \sum m_i$, we have that a^n is a sum of elements of the form $g_1^{i_1} \dots g_m^{i_m}$. Now because n is sufficiently large we have that, for any such product, $\sum i_j = n$ implying that at least one $i_j \geq m_j$. This implies that $g_j^{m_j} (g_1^{i_1} \dots g_j^{i_j - m_j} \dots g_m^{i_m}) \in J$, as J is an ideal. Therefore every product of the form $g_1^{i_1} \dots g_m^{i_m}$ such that $\sum i_j = n$ is in J , implying that $I^n \subseteq J$. $\triangle$

2.42: Homomorphisms of quotients by ideals containing one another.

Proof. Part (a): If $I \subseteq J$, then R/I has a natural map onto R/J given by the identity map defined on cosets of I . For $a, b \in R$, $a \sim b$ in R/I if and only if $a - b \in I$ which implies that $a - b \in J$ so $a \sim b$ in R/J . Then the fact that we have a homomorphism follows by the fact that R/J is a ring.

Part (b): If $R \subseteq S$ is a subring then we have the natural homomorphism $R/I \rightarrow R/IS$ by the fact that $I \subseteq IS$. Now there is another natural injective homomorphism $R/IS \rightarrow S/IS$ given by the identity, and the composition of these two gives the desired natural map. $\triangle$

2.43: The maximal ideal of a local ring of a point and ideals in the polynomial ring.

Proof. The maximal ideal of a point is given by all non-units of $\mathcal{O}_P$, i.e. those vanishing at P . A polynomial with no constant term must vanish at P , and I consists of precisely these polynomials. Therefore and $f \in I\mathcal{O}$ has numerator vanishing at P , implying that $f \in \mathfrak{m}$. In fact all these statements are “if-and-only-if,” so the result holds. $\triangle$

2.44: Quotients of defining ideals in the local ring of a point.

Proof. **Incomplete** $\triangle$

2.45: Ideals are comaximal if their varieties are disjoint.

Proof. Observe: $V(I) \cap V(J) = V(I + J) = \emptyset$ if and only if $I + J = R$. $\triangle$

2.46: Dimension of bivariate polynomials of degree smaller than n .

Proof. The power ideal I^n consists of all degree n polynomials in X and Y . Then $k[X, Y]/I^n$ is the set of all degree $d < n$ polynomials, which has dimension $n(n+1)/2$. $\triangle$

2.9 Ideals with a Finite Number of Zeros

2.10 Quotient Modules and Exact Sequences

2.48: Kernel and image of module homomorphisms are modules.

Proof. The kernel and image are, in particular, subgroups for any group homomorphism, and the R -action obviously extends. To see exactness, note that the inclusion $\ker \varphi \subseteq M$ has image equal to the kernel of φ (big surprise). Therefore the sequence has $\text{im} \subseteq \ker \varphi$. $\triangle$

2.49: Morphisms of modules.

Proof. Part (a): Consider the map $\bar{\varphi}$ that takes cosets $a + N \in M/N$ to elements $\varphi(a) \in M'$. This is well-defined because $\varphi(N) = 0$. Then $\bar{\varphi} \circ \pi(a) = \bar{\varphi}(a + N) = \varphi(a)$. Uniqueness follows by construction: Any other map factoring φ must have that $\bar{\varphi}(\pi(a)) = \varphi(a)$, so in particular cosets of a must go to $\varphi(a)$.

Part (b): If $P \subseteq N \subseteq M$ our natural homomorphisms are achieved by the same argument as in 2.42. Now

$$0 \rightarrow N/P \xrightarrow{\subseteq} M/P \xrightarrow{\varphi} M/N \rightarrow 0$$

has $\text{im } \varphi = M/N$. Then $\ker \varphi = \{a \in M/P \mid a \in N\} = \{a \in N/P\} = \text{im } \subseteq$, so the sequence is exact.

Part (c): This is a consequence of (b) when $P \subseteq N \subseteq M$ are finite-dimensional vector spaces and an application of the rank-nullity theorem.

Part (d): Ideals are obviously modules. Then this follows from (b) taking $I = P \subseteq J = N \subseteq R = M$.

Part (e): This is (d) with $I = \mathfrak{m}^n$, $J = \mathfrak{m}^{n+1}$, and $R = \mathcal{O}$. $\triangle$

2.50: DVRs, maximal ideals, and modules.

Proof. We suppose we have a DVR R with maximal ideal $\mathfrak{m}$, quotient field K , and for k a subring of R we have an isomorphism $k \rightarrow R \rightarrow R/\mathfrak{m}$.

Part (a): Consider the short exact sequence

$$0 \rightarrow k \rightarrow R \rightarrow R/\mathfrak{m} \rightarrow 0.$$

Then a natural sequence of R -modules follows by 2.49(e):

$$0 \rightarrow \mathfrak{m}$$

$\triangle$

2.11 Free Modules